

No. of Functional Features Included in the Solution

Date	08 October 2026
Team ID	SWTID-2026-2386
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Ask a question	Educational answer to any academic question	qna.py, /qa	Done	Core
2	Follow-up questions	The previous question and answer are sent as context for the next question	qna.py, app.js	Done	Core
3	Concept explanation	Definition, key ideas, example and common mistake	explanation_module.py, /explain	Done	Core
4	Summarization	Key points from pasted study material	summary_module.py, /summarize	Done	Core
5	Quiz generation	Three multiple-choice questions with four options; JSON cleaned and validated	quiz_module.py, /quiz	Done	Core
6	Quiz feedback	Chosen option turns green or red and the correct answer is shown	app.js, style.css	Done	Core
7	Learning path	Beginner to advanced stages with time estimates and resources	learning_path.py, /learn/recommendations	Done	Core
8	Input validation	Empty, too long (over 8000 characters) and malformed requests are rejected	main.py, models.py	Done	Core
9	Friendly error handling	Missing key, rejected key, rate limit, network failure, empty or invalid Gemini reply	gemini_utils.py, main.py	Done	Core
10	Gemini model fallback	Tries other models if the chosen model is retired or busy	gemini_utils.py, config.py	Done	Additional
11	API key protection	Key only in .env, read on the server, .env ignored by git	config.py, .gitignore	Done	Core
12	Health check and Docker	/health endpoint and a Dockerfile	main.py, Dockerfile	Done	Additional

Feature Summary

Metric	Count / Value
Total Features Planned	12
Total Features Implemented	12
Core / Must-Have Features	10
Additional / Nice-to-Have Features	2
Features Tested & Verified	Automated tests exist for every feature (tests/test_app.py); the team records the run in 6.Project Testing/evidence/pytest_results.txt

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Follow-up questions, quiz feedback
2	Backend / Logic	4	Question answering, explanation, summary, learning path
3	Database / Storage	0	Not used (stateless)
4	API / Integration	3	Quiz generation (Gemini JSON), model fallback, health check and Docker
5	Security / Authentication	3	Input validation, friendly error handling, API key protection